

Yue-Hong Chen

Ph.D. Candidate in Astronomy

Nanjing University, Nanjing, China
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Research Interests

My research bridges solar and stellar physics. Starting from observational studies of solar eruptive activity, I have extended my work to high-resolution magnetohydrodynamic (MHD) simulations of Sun-like stars, covering the full chain from stellar coronae and winds to coronal mass ejections and the associated energetic particle events.

Education

- Sep 2023 – **Ph.D. in Astronomy**, *Nanjing University*, Nanjing, China.
Present Advisor: Prof. Xin Cheng.
Research focus: high-resolution MHD modelling of stellar coronae, winds, and CMEs in Sun-like stars.
- Sep 2025 – **Visiting Ph.D. Researcher**, *Leibniz Institute for Astrophysics Potsdam (AIP)*, Potsdam, Germany.
Present Host: Dr. Julián D. Alvarado-Gómez.
Research focus: stellar wind and CME modelling of Sun-like stars at varying rotation rates.
- Sep 2021 – **M.Sc. in Astronomy**, *Nanjing University*, Nanjing, China.
Jun 2023 Advisor: Assoc. Prof. Yu Dai.
Research focus: observational study of solar flares and filament eruptions.

Additional Research Visits

- Sep – Nov 2024 **Visiting Researcher**, *Leibniz Institute for Astrophysics Potsdam (AIP)*, Potsdam, Germany.
- Aug 2023 **Joint-Training Visiting Student**, *Yunnan University*, Kunming, China.

Publications

Refereed Journal Articles

- 2025 **Chen, Y.-H.**, Alvarado-Gómez, J. D., Cheng, X., Dai, Y., Shi, T., Poppenhäger, K., Xing, C., Inoue, S., Warnecke, J., Korpi-Lagg, M. J., & Ding, M. *High-Resolution Modeling of Coronae and Winds in Solar-Type Stars with Varying Rotation Rates. I. X-Ray Coronae*. *ApJ*, 995, 83. [doi:10.3847/1538-4357/ae1697](https://doi.org/10.3847/1538-4357/ae1697)
- 2023 **Chen, Y.**, Cheng, X., Chen, J., Dai, Y., & Ding, M. *Observations of a Failed Solar Filament Eruption Involving External Reconnection*. *ApJ*, 959, 67. [doi:10.3847/1538-4357/ad09d8](https://doi.org/10.3847/1538-4357/ad09d8)
- 2023 **Chen, Y.**, Dai, Y., & Ding, M. *An Atypical Plateau-Like Extreme-Ultraviolet Late-Phase Solar Flare Driven by the Nonradial Eruption of a Magnetic Flux Rope*. *A&A*, 675, A147. [doi:10.1051/0004-6361/202345914](https://doi.org/10.1051/0004-6361/202345914)

Submitted / Under Review

Chen, Y.-H., Alvarado-Gómez, J. D., et al., *High-Resolution Modelling of Coronae and Winds in Solar-type Stars with Varying Rotation Rates. II. Stellar Winds*, under review.

Conference Presentations (Oral)

- Jun 2026 Cool Stars 23, Tokyo, Japan. (*Awarded Best Poster, granted oral presentation opportunity.*)
- Mar 2026 89th DPG Annual Meeting & SMuK Spring Conference, Extraterrestrial Physics (EP) Division. FAU Erlangen-Nürnberg, Germany.

- Aug 2025 16th Zhang Heng Symposium / Joint Solar–Stellar Physics Workshop (Centenary of Chinese Hand-Drawn Sunspot Records). Qingdao, China.
- Jul 2025 IAU Symposium 400, “Solar and Stellar Multi-Scale Activity.” EAFIT University, Medellín, Colombia.
- May 2025 “Stellar X-Ray Flares in the Einstein Probe Era” Workshop (EP Science Center & NAOC Solar Magnetic Activity Group). Beijing, China.
- Apr 2025 2nd International Conference of Deep Space Sciences (ICDSS 2025). Hefei, China. (*Best Presentation Award.*)
- Jul 2024 45th COSPAR Scientific Assembly. BEXCO, Busan, Korea.
- May 2024 IAU Symposium 388, “Solar and Stellar Coronal Mass Ejections.” Jagiellonian University Observatory, Kraków, Poland.
- 2023 Annual Meeting of the Chinese Solar Physics Community. Chengdu, China.

Grants & Funding

- Apr 2024 – **Formation and Evolution of Stellar Coronal Mass Ejections.** *Principal Investigator.*
- May 2026 Postgraduate Research & Practice Innovation Program of Jiangsu Province.
- Sep 2024 – **Lower-Atmospheric Signatures of Coronal Mass Ejection Initiation.** *Co-Investigator*
- Aug 2027 (numerical simulation). Natural Science Foundation of Jiangsu Province (Youth Program).
PI: Dr. Chen Xing.

Teaching

- TA *Introduction to Space Physics* (undergraduate course), Nanjing University.

Skills

- Programming Python, IDL, Fortran
- Simulation MPI-AMRVAC, SWMF
codes
- Methods MHD simulation; solar and stellar data analysis

Languages

- Mandarin Native
Chinese
- English Working proficiency

References

- Ph.D. advisor Prof. Xin Cheng, School of Astronomy and Space Science, Nanjing University.
- Visit host Dr. Julián D. Alvarado-Gómez, Leibniz Institute for Astrophysics Potsdam (AIP).
- M.Sc. advisor Assoc. Prof. Yu Dai, School of Astronomy and Space Science, Nanjing University.